

CS5340 – Uncertainty Modeling in AI
(Semester 1 : AY2024/25)

Time Allowed : 2 Hours

INSTRUCTIONS TO STUDENTS

1. Please write your Student Number only. Do not write your name.
2. This assessment paper contains **FOUR** questions and comprises **FIVE** printed pages.
3. Students are required to answer **ALL** questions.
4. Students should write the answers for each question on a new page.
5. This is a **CLOSED BOOK** assessment.
6. Students can bring in **ONE A4 summary sheet**.
7. Non-programmable electronic calculators are allowed.

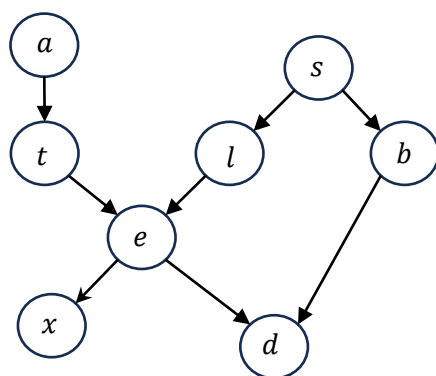
Question 1

a) **Prove** the following statements to be either true or false:

- i. $X \perp \{A, B\} \Rightarrow X \perp A$ and $X \perp B$
- ii. $X \perp \{A, B\} \Leftrightarrow X \perp A \mid B$ and $X \perp B \mid A$
- iii. $X \perp A \mid B$ and $X \perp B \Leftrightarrow X \perp \{A, B\}$
- iv. $X \perp Y \mid \{W, Z\}$ and $X \perp W \mid \{Y, Z\} \Leftrightarrow X \perp \{Y, W\} \mid Z$

(17 marks)

b) Figure 1.1 shows the Chest Clinic network which concerns the diagnosis of lung disease (Tuberculosis, lung cancer, or both, or neither). In this model, a visit to Asia is assumed to increase the probability of tuberculosis.



x :	Positive X-ray
d :	Dyspnea (Shortness of Breath)
e :	Either tuberculosis or lung cancer
t :	Tuberculosis
l :	Lung Cancer
b :	Bronchitis
a :	Visited Asia
s :	Smoker

Fig. 1.1: Belief Network for Chest Clinic

State **with reasoning** if the following conditional independence relations are true or false:

- i. Tuberculosis $\perp$ Smoking $\mid$ Shortness of Breath
- ii. Lung Cancer $\perp$ Bronchitis $\mid$ Smoking
- iii. Visit to Asia $\perp$ Smoking $\mid$ Lung Cancer
- iv. Visit to Asia $\perp$ Smoking $\mid$ Lung Cancer, Shortness of Breath

(8 marks)

Show all your workings clearly.

Question 2

We are given two biased coins, Coin A and Coin B, each with an unknown probability of landing heads. In a series of trials, we randomly choose one of the two coins and flip it a single time. However, we **do not know** which coin was chosen for each flip.

Let us denote the random variables of choosing each coin as a 1-of-2 representation $Z_k \in \{0,1\}$, where $Z_{k=0} = 1 := \text{Coin A}$ and $Z_{k=1} = 1 := \text{Coin B}$, and the outcome of a coin flip as $X \in \{0,1\}$, where $X = 0 := \text{Head}$ and $X = 1 := \text{Tail}$. We further denote the probabilities of choosing Coin B as π , and getting tail for Coin A and Coin B as μ_0 and μ_1 , respectively.

- a) Given N *i.i.d* observations of the coin flips, draw the Bayesian network and write the joint probability distribution that represent the coin choosing and flipping process.
(10 marks)
- b) Using the **EM algorithm**, derive the expressions of μ_0 and μ_1 after the Maximization step.
(10 marks)
- c) The results of these flips are recorded as a sequence of heads (H) and tails (T): $\{H, T, H\}$. Taking equal probability of choosing either Coin A or B and the initial values of $\mu_0 = 0.6$ and $\mu_1 = 0.5$, perform one iteration of the EM algorithm to update μ_0 and μ_1 .
(5 marks)

Show all your workings clearly.

Question 3

- a) Let $Y = \{Y_i\}_{i=1}^N$ denote a noisy binary image with N pixels, where each pixel Y_i takes a value of 0 (black) or 1 (white). We want to infer the true underlying image $X = \{X_i\}_{i=1}^N$, where each X_i is also binary. The denoising model assumes the following:

Prior on X : Neighboring pixels in the true image X are likely to have the same value, encouraging spatial smoothness, as represented as a pairwise Markov random field prior:

$$p(X) \propto \exp(\beta \sum_{(i,j) \in \text{neighbors}} X_i X_j),$$

where $\beta > 0$ is a parameter that controls the strength of the smoothness prior and “neighbors” denote all neighbors of each node X_i in the Markov random field.

Observation Model: The probability of observing $Y_i = 1$ given $X_i = 1$ is λ , and the probability of observing $Y_i = 0$ given $X_i = 0$ is also λ , where $0.5 < \lambda \leq 1$ to encourage Y_i in taking the same label as the observation.

Using **variational inference with a mean-field approximation** to approximate the posterior distribution $p(X | Y)$:

- i. Write the conditional probability distribution for the observation model. (2 marks)
 - ii. Derive the approximate marginal posterior and the mean value of each pixel i . (8 marks)
- b) Figure 3.1 shows the Bayesian network of a **three-component mixture of exponential distribution**. $p(Y | \pi) = \prod_k \pi_k^{Y_k}$, where $Y_k \in \{0,1\}$ is a binary random variable with 1-of-3 representation, and the parameters $\pi = \{\pi_1 = 0.5, \pi_2 = 0.3, \pi_3 = 0.2\}$. $X \in \mathbb{R}_{\geq 0}$ and $p(X | Y, \lambda) = \prod_k (\lambda_k \exp\{-\lambda_k X\})^{Y_k}$, where the parameters $\lambda = \{\lambda_1 = 0.5, \lambda_2 = 1.0, \lambda_3 = 1.5\}$.

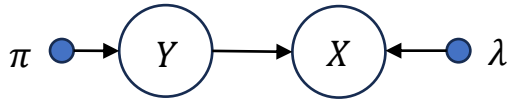


Fig. 3.1: Mixture of exponential distribution

- i. Given the following 5 pairs of random numbers drawn from a uniform distribution $\text{unif}(0,1)$:

$$\{\{0.76, 0.82\}, \{0.74, 0.69\}, \{0.39, 0.32\}, \{0.66, 0.95\}, \{0.17, 0.03\}\},$$

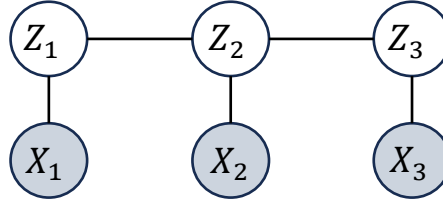
use **ancestral sampling** to draw 5 sets of samples of (X, Y) . We are further given the following useful equation: $\int a \exp\{at\} dt = \exp\{at\} + \text{const}$.

(10 marks)

- ii. Using the drawn samples, find the conditional probability of $p(Y | 0 < X \leq 1)$.

(5 marks)

Show all your workings clearly.

Question 4**Fig. 4.1: Markov Random Field****Table 4.1: Energy Term of Likelihood**

X_i	Z_i	$E(Z_i, X_i)$
0	0	2
0	1	6
1	0	6
1	1	2

Table 4.2: Energy Term of Prior

Z_i	Z_j	$E(Z_i, Z_j)$
0	0	1
0	1	4
1	0	4
1	1	1

Figure 4.1 shows a Markov random field (MRF) with three binary latent random variables $Z = \{Z_1, Z_2, Z_3\} \in \{0,1\}$ and three binary observed random variables $X = \{X_1 = 0, X_2 = 1, X_3 = 0\}$. The joint probability follows the Gibbs distribution:

$$p(Y) = \frac{1}{Z_p} \exp\{-\sum_{c \in C} E(Y_c)\},$$

where C denotes all cliques in the MRF and Z_p is the partition function. Table 4.1 and 4.2 gives the energies of the likelihood $E(Z_i, X_i)$ and priors $E(Z_i, Z_j)$, where i and j are the neighbors in the MRF.

- Using the **graph-cut algorithm**, solve for **all configurations** of the latent random variables Z for the maximum a posteriori (MAP) probability. Show all details in forming the flow-graph and max-flow min-cut. (8 marks)
- Using the **max-product algorithm**, solve for **all configurations** of the latent random variables Z for the MAP probability. (8 marks)
- List exhaustively** the energies that correspond to each configuration of the latent random variables Z . Compare this answer with those in a) and b). (7 marks)
- State the configuration of Z that gives the **minimum** MAP probability and **explain why** this configuration leads to the minimum MAP in terms of the prior and likelihood. (2 marks)

Show all your workings clearly.

--End--